

Recap: scaling in the wild

● what are challenges in scaling 'in practice'?

- ①: setting model arch hyperparameters (width, etc)
- ②: setting optimizer hyperparameters (LR, Batch size)
- ③: compute needed to fit the big chinchilla sweep

some solutions:

- 1: Assume stability (or use map)
- 2: Search for optimal LR / batch in small scale, either keep fixed or predict scaling.
- 3: use alternative learning rate schedule, like WSD: step-wise decay chinchilla

Lecture #12. Model Evaluation

given a model, how good is it?

Benchmark Scores: ^{↑ MMLU} MMLU, Math, ^{↑ LiveCode Bench} Codeforces, [↑] SWB-Bench

↓

what are these benchmarks?

Q/A truth.

pop art

what do these numbers mean?

Model ranking: Leaderboard Chatbot Arena

what is the point of evaluation? there is no one true evaluation: it depends on what question you're trying to answer.
^{what role/organization}

(1) user or company wants to make a purchase decision

(model A versus model B) for their use case (e.g. customer service chatbots).

(2) Researchers want to measure the raw capabilities of a model.
(e.g. scientific discovery)

(3) governments/public sectors: we want to understand the benefits + harms of a model (for business and policy reasons)

(4) Model developers need feedback to improve the model

Evaluation Frameworks:

1. what are the inputs?

3. How do you evaluate the outputs?

2. how do call the LLM?

4. How to interpret results?